

PROGRAM - 10

TITLE: Demonstrate kubernetes dashboard and CLI for cluster monitoring

Program Structure

Program-10

- service.yaml
- deployment.yaml

Step 1: Start the minikube with specified cpus and memory

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-10$ minikube start --cpus=2
minikube v1.37.0 on Ubuntu 22.04
Using the docker driver based on existing profile
Starting "minikube" primary control-plane node in "minikube" cluster
Pulling base image v0.0.48 ...
Restarting existing docker container for "minikube" ...

🐳 Docker is nearly out of disk space, which may cause deployments to fail! (95% of capacity). You can pass '--force' to skip this check.
Suggestion:

Try one or more of the following to free up space on the device:

1. Run "docker system prune" to remove unused Docker data (optionally with "-a")
2. Increase the storage allocated to Docker for Desktop by clicking on:
   Docker icon > Preferences > Resources > Disk Image Size
3. Run "minikube ssh -- docker system prune" if using the Docker container runtime
Related issue: https://github.com/kubernetes/minikube/issues/9024

🔧 Preparing Kubernetes v1.34.0 on Docker 28.4.0 ...
Verifying Kubernetes components...
   ■ Using image gcr.io/k8s-minikube/storage-provisioner:v5
   ■ Using image docker.io/kubernetesui/metrics-scraper:v1.0.8
   ■ Using image registry.k8s.io/metrics-server/metrics-server:v0.8.0
   ■ Using image docker.io/kubernetesui/dashboard:v2.7.0
💡 Some dashboard features require the metrics-server addon. To enable all features please run:

    minikube addons enable metrics-server

🌟 Enabled addons: metrics-server, storage-provisioner, dashboard, default-storageclass

! /usr/bin/kubectl is version 1.28.15, which may have incompatibilities with Kubernetes 1.34.0.
   ■ Want kubectl v1.34.0? Try 'minikube kubectl -- get pods -A'
🏠 Done! kubectl is now configured to use "minikube" cluster and "default" namespace by default
```

Step 2: Enable the minikube dashboard

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-10$ minikube dashboard
Verifying dashboard health ...
Launching proxy ...
Verifying proxy health ...
Opening http://127.0.0.1:40167/api/v1/namespaces/kubernetes-dashboard/services/http:kubernetes-dashboard:/proxy/ in your default browser..
Opening in existing browser session.
```

Step 3: Create a service.yaml file

```
GNU nano 6.2
apiVersion: v1
kind: Service
metadata:
  name: nginx-service
spec:
  selector:
    app: nginx
  ports:
    - protocol: TCP
      port: 80
      targetPort: 80
  type: NodePort
```

Step 4: Create a deployment.yaml file

```
GNU nano 6.2
apiVersion: apps/v1
kind: Deployment
metadata:
  name: nginx-deployment
  labels:
    app: nginx
spec:
  replicas: 2
  selector:
    matchLabels:
      app: nginx
  template:
    metadata:
      labels:
        app: nginx
    spec:
      containers:
      - name: nginx
        image: nginx:latest
        ports:
        - containerPort: 80
```

Step 5: Apply deployment and service.yaml file

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-10$ kubectl apply -f deployment.yaml
deployment.apps/nginx-deployment created
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-10$ kubectl apply -f service.yaml
service/nginx-service created
```

Step 6: Monitor via CLI

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-10$ kubectl get pods
NAME                                READY   STATUS    RESTARTS   AGE
nginx-deployment-6f9664446b-hmz64   1/1     Running   0           47s
nginx-deployment-6f9664446b-rjhhc   1/1     Running   0           47s
```

Step 7: Check nodes

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-10$ kubectl get nodes
NAME       STATUS   ROLES    AGE   VERSION
minikube   Ready    control-plane  18d   v1.34.0
```

Step 8: Check Pod logs

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-10$ kubectl logs nginx-deployment-6f9664446b-hmz64
/docker-entrypoint.sh: /docker-entrypoint.d/ is not empty, will attempt to perform configuration
/docker-entrypoint.sh: Looking for shell scripts in /docker-entrypoint.d/
/docker-entrypoint.sh: Launching /docker-entrypoint.d/10-listen-on-ipv6-by-default.sh
10-listen-on-ipv6-by-default.sh: info: Getting the checksum of /etc/nginx/conf.d/default.conf
10-listen-on-ipv6-by-default.sh: info: Enabled listen on IPv6 in /etc/nginx/conf.d/default.conf
/docker-entrypoint.sh: Sourcing /docker-entrypoint.d/15-local-resolvers.envsh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/20-envsubst-on-templates.sh
/docker-entrypoint.sh: Launching /docker-entrypoint.d/30-tune-worker-processes.sh
/docker-entrypoint.sh: Configuration complete; ready for start up
2025/12/19 04:33:34 [notice] 1#1: using the "epoll" event method
2025/12/19 04:33:34 [notice] 1#1: nginx/1.29.4
2025/12/19 04:33:34 [notice] 1#1: built by gcc 14.2.0 (Debian 14.2.0-19)
2025/12/19 04:33:34 [notice] 1#1: OS: Linux 6.8.0-87-generic
2025/12/19 04:33:34 [notice] 1#1: getrlimit(RLIMIT_NOFILE): 1048576:1048576
2025/12/19 04:33:34 [notice] 1#1: start worker processes
2025/12/19 04:33:34 [notice] 1#1: start worker process 29
2025/12/19 04:33:34 [notice] 1#1: start worker process 30
2025/12/19 04:33:34 [notice] 1#1: start worker process 31
2025/12/19 04:33:34 [notice] 1#1: start worker process 32
```

Step 9: Describe the pod

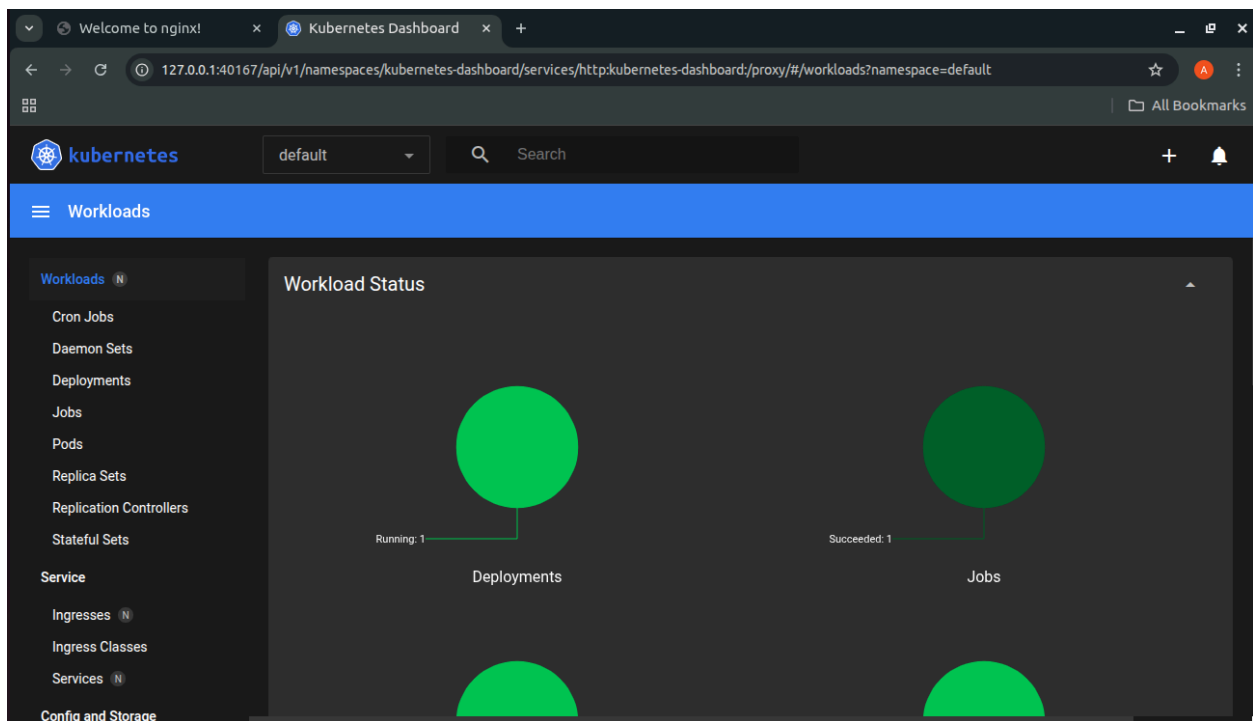
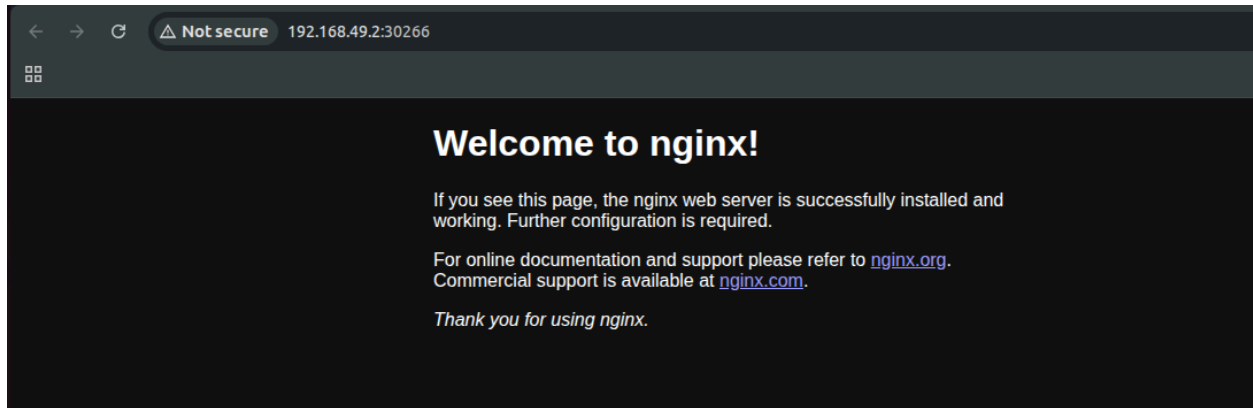
```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-10$ kubectl describe pods nginx-deployment-6f9664446b-hmz64
Name:          nginx-deployment-6f9664446b-hmz64
Namespace:     default
Priority:       0
Service Account: default
Node:          minikube/192.168.49.2
Start Time:    Fri, 19 Dec 2025 10:03:27 +0530
Labels:        app=nginx
               pod-template-hash=6f9664446b
Annotations:   <none>
Status:        Running
IP:            10.244.1.34
IPs:
  IP:          10.244.1.34
Controlled By: ReplicaSet/nginx-deployment-6f9664446b
Containers:
  nginx:
    Container ID:  docker://d6ff8eea26f98b28f0229442751a8a93f63e1c9268f88f3d6de6d33a52ca6c99
    Image:         nginx:latest
    Image ID:      docker-pullable://nginx@sha256:fb01117203ff38c2f9af91db1a7409459182a37c87cced5cb442d1d8fcc66d19
    Port:         80/TCP
    Host Port:    0/TCP
    State:        Running
      Started:    Fri, 19 Dec 2025 10:03:34 +0530
    Ready:        True
    Restart Count: 0
    Environment:  <none>
    Mounts:
      /var/run/secrets/kubernetes.io/serviceaccount from kube-api-access-npx9r (ro)
Conditions:
  Type                                Status
  PodReadyToStartContainers           True
  Initialized                          True
  Ready                               True
  ContainersReady                     True
  PodScheduled                        True
```

Step 10: Access the application via Minikube and Viewing the same in the web browser

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/Labprograms/Program-10$ minikube service nginx-service
```

NAMESPACE	NAME	TARGET PORT	URL
default	nginx-service	80	http://192.168.49.2:30266

 Opening service default/nginx-service in default browser...



Step 12: Clean up process

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-10$ kubectl delete pods --all
pod "nginx-deployment-6f9664446b-hmz64" deleted
pod "nginx-deployment-6f9664446b-rjhhc" deleted
```

Step 13: Delete all the services

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-10$ kubectl delete svc --all
service "kubernetes" deleted
service "nginx-service" deleted
```

Step 14: Delete all the deployments

```
1RV24MC018_Ananya_h@user-ThinkPad-E480:~/labprograms/Program-10$ kubectl delete deployments --all  
deployment.apps "nginx-deployment" deleted
```